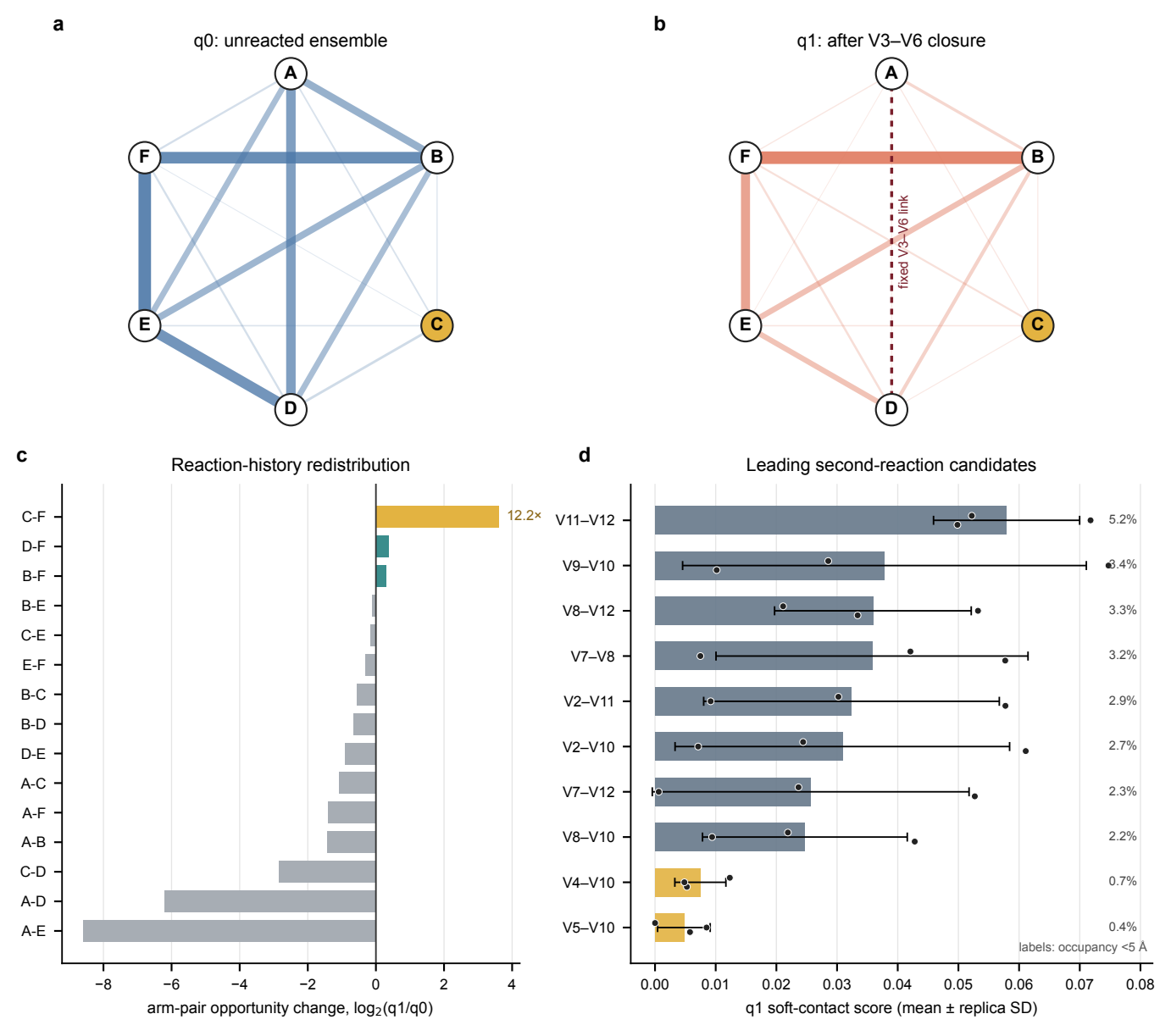
a
q0: unreacted ensemble
A
B
C
D
E
F
b
q1: after V3–V6 closure
fixed V3–V6 link
c
Reaction-history redistribution
C-F
D-F
B-F
B-E
C-E
E-F
B-C
B-D
D-E
A-C
A-F
A-B
C-D
A-D
A-E
12.2×
−8
−6
−4
−2
0
2
4
arm-pair opportunity change, log2(q1/q0)
d
Leading second-reaction candidates
V11–V12
V9–V10
V8–V12
V7–V8
V2–V11
V2–V10
V7–V12
V8–V10
V4–V10
V5–V10
5.2%
3.4%
3.3%
3.2%
2.9%
2.7%
2.3%
2.2%
0.7%
0.4%
labels: occupancy <5 Å
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
q1 soft-contact score (mean ± replica SD)